

MA1521 Calculus for Computing

Lecture Group 2

Lecturer: Prof To Wing Keung

Lesson 08: Friday 04 September 2026

(Week 4)

Remark. Q07.6 to Q07.9 are leftover exercises from the Tuesday class in Week 4.

Remark. Q08.1 to Q08.8 are on Section 13.7 of Thomas' Calculus (and Lecture 9 and Lecture 10 of Prof Auroux).

Q07.6. (i) Suppose $f(\mathbf{x}, \mathbf{y})$ is a function such that

$$f_x = 3x^2 + Axy, \quad f_y = 3x^2 + 6y^2,$$

where A is a constant. Using Clairaut's Theorem (i.e. the identity $f_{xy} = f_{yx}$), determine the value of A .

(ii) Using your result in (i), find such a function $f(\mathbf{x}, \mathbf{y})$ by inspection or any method using your experience in one variable calculus.

[Remark. Such a function is not unique.]

Q07.7. Let $f(\mathbf{x}, \mathbf{y})$ be a function, and $(\mathbf{x}_o, \mathbf{y}_o)$ be a point in the domain of f . Consider the linear polynomial (in two variables) given by

$$\mathbf{g}(\mathbf{x}, \mathbf{y}) = \mathbf{a} + \mathbf{b}(\mathbf{x} - \mathbf{x}_o) + \mathbf{c}(\mathbf{y} - \mathbf{y}_o)$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are constants.

(i) Determine the values of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ (in terms of data on f) if

$$\mathbf{g}(\mathbf{x}_o, \mathbf{y}_o) = f(\mathbf{x}_o, \mathbf{y}_o), \quad \mathbf{g}_x(\mathbf{x}_o, \mathbf{y}_o) = f_x(\mathbf{x}_o, \mathbf{y}_o)$$

$$\text{and} \quad \mathbf{g}_y(\mathbf{x}_o, \mathbf{y}_o) = f_y(\mathbf{x}_o, \mathbf{y}_o).$$

(ii) Write down the equation for the graph of $\mathbf{g}(\mathbf{x}, \mathbf{y})$.

[Remark: Note that the equation in (ii) is simply an equation of the tangent plane to the graph of $f(\mathbf{x}, \mathbf{y})$ at the point $(\mathbf{x}_o, \mathbf{y}_o)$ as given in Prof Auroux's notes, Lecture 9.]

Q07.8. Consider the function $f(x, y) = x^2y + 2xy$. Find an equation of the tangent plane to the graph of $f(x, y)$ at the point $(1, 2)$. [Recall the result in **Q07.7**.]

Q07.9. Consider the surface in $\mathbb{R}^3$ defined by the equation $yz = x^2$. Find an equation of the tangent plane to the surface at the point $(4, 2, 8)$.

[Suggestion: First rewrite the surface as the graph of a function of two variables.]

Q08.1. Find the critical points of the function

$$f(x, y) = 2xy - x^2 - 2y^2 + 3x.$$

Q08.2. Determine all the critical points of the function

$$f(x, y) = 2xy - x^2y - xy^2,$$

and classify each critical point as a local maximum point, a local minimum point or a saddle point of f . Evaluate also the local maximum values and local minimum values of f , if any.

[Suggestion: You may find the second derivative test useful.]

Q08.3. Determine all the critical points of the function

$$f(x, y) = \ln(xy^2) - xy - 2y$$

over the region $x > 0$, $y > 0$ in the xy -plane. Classify each critical point as a local maximum point, a local minimum point or a saddle point of f . Evaluate also the local maximum values and local minimum values of f , if any.

[Suggestion: You may find the second derivative test useful.]

Q08.4. Consider the surface $y^2 - xz = 4$ in the xyz -space. Find the point(s) on the surface which is closest to the origin $(0, 0, 0)$.

[Suggestion: First formulate the question as a problem to minimize a certain function in the variables x, z .]

Q08.5. A rectangular box with its top left open has its five faces made from different materials. The front face and the back face are made from materials costing 10 dollars per square meter. The two lateral faces are made from materials costing 5 dollars per square meter. The bottom face is made from materials costing 30 dollars per square meter. Find the minimum material cost of the rectangular box if the volume of the box is 6 cubic meters. Give also the dimensions (i.e. length, width and height) of such box.

Q08.6. Find the maximum and minimum points of the function $f(x, y) = x^2 - 2xy + 2y^2 - 2y$ over the rectangular region $R : -1 \leq x \leq 2, 0 \leq y \leq 2$ in the xy -plane. Hence find the maximum and minimum values of f over the region R .

[Remark: Here you are asked to find the absolute/global maxima and minima of the function f over the region R .]

Q08.7. Using the method of least squares, find the line (of the form $y = ax + b$) which best fits the three data points $(-1, 1)$, $(1, 2)$ and $(3, 2)$.

Q08.8. Using the method of least squares, find the quadratic polynomial of the form $y = ax^2 + bx$ which best fits the three data points $(-1, 1)$, $(1, 1)$ and $(2, 0)$.